

Problem Set 6

1

Some hackers have overclocked the smart toilet's flush for extreme power! As a result, the motor now emits energy at microwave frequencies (2.45 GHz), which could be harmful.

You are asked to update the product safety manual to tell these users how to remotely flush their toilet from a safe distance using a Wi-Fi enabled wall switch.

Assume the safe power density for human exposure is 1 mW/cm^2 at this frequency. Assume R is the distance in meters between the user and the toilet. If the flush motor emits a wave with an electric field amplitude that decays as $E(R) = 150/R \text{ V/m}$ how far away should the user stand when operating the flush feature?

✓ Answer ✓

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 376.73031341180507$$

$$S = \frac{|E|^2}{2\eta_0} = \frac{\left(\frac{150}{R}\right)^2}{2\eta_0}$$

$$10 \text{ W/m} = \frac{\left(\frac{150}{R}\right)^2}{2\eta_0}$$

$$R = 1.7280685955753878 \text{ m}$$

The users should stand at least 1.73 m away.

7.3

The electric field phasor of a uniform plane wave is given by $\tilde{\mathbf{E}} = \hat{\mathbf{y}}10e^{j0.2z} \text{ V/m}$. If the phase velocity of the wave is $1.5 \times 10^8 \text{ m/s}$ and the relative permeability of the medium is $\mu_r = 2.4$, find:

a

the wavelength

✓ Answer

$$\beta = -0.2$$

$$\lambda = \frac{2\pi}{|\beta|} = 31.41592653589793 \text{ m}$$

b

the frequency f of the wave

✓ Answer

$$f = \frac{u_p}{\lambda} = 4.77464829275686 \text{ MHz}$$

c

the relative permittivity of the medium

✓ **Answer**

$$u_p = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$$
$$\epsilon_r = \frac{1}{u_p^2 \mu_0 \epsilon_0 \mu_r} = 1.6643614421032316$$

d

the magnetic field $H(z, t)$

✓ **Answer**

$$H = \frac{1}{\eta} (\hat{v} \times E)$$
$$H = 0.02210485321012625 \cos(2\pi(4774648.29275686)t + 0.2z) \text{ A/m}$$

7.17

For each of the following combinations of parameters, determine if the material is a low-loss dielectric, a quasiconductor, or a good conductor. Then calculate $\alpha, \beta, \lambda, \mu_p, \eta_c$.

1. Glass with $\mu_r = 1, \epsilon_r = 5, \sigma = 10^{-12} \text{ S/m}$ at 10 GHz.
2. Animal tissue with $\mu_r = 1, \epsilon_r = 12, \sigma = 0.3 \text{ S/m}$ at 100 MHz.
3. Wood with $\mu_r = 1, \epsilon_r = 3, \sigma = 10^{-4} \text{ S/m}$ at 1 kHz.

✓ **Answer**

Parameter	(1) Glass	(2) Animal Tissue	(3) Wood
L	3.6×10^{-13}	4.49	599.17
Class	Low-loss dielectric	Quasiconductor	Good conductor
α (Np/m)	8.42×10^{-11}	9.75	6.28×10^{-4}
β (rad/m)	468.65	12.15	6.29×10^{-4}
λ (m)	0.0134	0.517	9991.66
u_p (m/s)	1.34×10^8	5.17×10^7	9.99×10^6
η_c (Ω)	$168.48 + j0$	$39.54 + j31.71$	$6.29 + j6.28$

Via a python script.

7.23

The skin depth of a certain nonmagnetic conducting material is $3 \mu\text{m}$ at 5 GHz. Determine the phase velocity in the material.

✓ Answer

$$\delta_s = \frac{1}{\alpha} = \frac{1}{\beta}$$

$$\omega = 2\pi f$$

$$u_p = \frac{\omega}{\beta} = \omega \delta_s = 2\pi f \delta_s = 94247.7796076938 \text{ m/s}$$

7.27

At 2 GHz, the conductivity of meat is on the order of 1 S/m. When a material is placed inside a microwave oven and the field is activated, the presence of the electromagnetic fields in the conducting material causes energy dissipation in the material in the form of heat.

a

Develop an expression for the time-average power per mm^3 dissipated in a material of conductivity σ if the peak electric field in the material is E_0 .

✓ Answer

$$P = VI$$

$$V = Ed$$

$$I = JA = \sigma EA$$

$$P = d\sigma E^2 A$$

$$\frac{P}{V} = \sigma E^2$$

$$P_{avg \text{ mm}^3} = \frac{\sigma E_0^2}{2}$$

b

Evaluate the result for an electric field $E_0 = 4 \times 10^4 \text{ V/m}$.

✓ Answer

$$P = 0.8 \text{ W/mm}^3$$

7.29

A rectangular copper block is 30 cm in height (along z). In response to a wave incident upon the block from above, a current is induced in the block in the positive x direction. Determine the ratio of the AC resistance of the block to its DC resistance at 1 kHz.

✓ Answer

$$\frac{R_{AC}}{R_{DC}} = \frac{\sigma A}{\sigma \omega \delta_s} = \frac{0.3}{\delta_s} = 0.3 \sqrt{\pi f \mu \sigma} = 143.55394103525572$$